Hardware Tools and Techniques

Objectives:

At the end of this episode, I will be able to:

1. Identify tools and techniques to troubleshoot hardware failures.

Exam Objective: *4.2 - Given a scenario, troubleshoot common hardware failures.*

Description: In this episode, the viewer will identify common component issue such as CPU, RAM, power and utilization as well as some of the tools and techniques performed to diagnose and resolve these issues.

- Hardware Components
  - System Issues
    - Firmware upgrades/downgrades
    - CMOS battery failure
    - Out-of-band management module failure
    - LED indicators, on-screen displays, LCD screens
    - Visual, auditory, smell (olfactory)
  - CPU issues
    - Cooling failure
    - Heatsink/fan/thermal component
    - Dust and debris
    - Thermal damage
    - Diagnostic tools
      - memtest86+ (checks cache)
      - Event logs
      - Software (HWMonitor)
      - Compressed air
  - RAM issues
    - Memory errors and failures
    - Failure
    - Incompatible
    - Improperly seating
    - Diagnostic tools
      - memtest86+/Windows Memory Diagnostic Tool
      - Multimeter
      - Web interfaces
      - Event logs
      - POST Errors
  - Power issues
    - PSU failure/fault
    - PDU/UPS/Circuit issues
    - Diagnostic tools
      - Multimeter
      - Circuit tester
      - Known good components
      - Compressed air
  - Utilization

        - Physical (VM host/on-prem)
            - Bottlenecks
            - Misconfiguration
            - Diagnostic tools
                - HWMonitor
                - Performance baselines
        - Cloud
            - Improper sizing
            - Misconfiguration
            - Lack of scalability options
            - Diagnostic tools:
                - Cloud-based dashboards
                - Performance monitoring
                - Elastic scalability features

- Demo
    - A walkthrough of HWMonitors component dianostics (CPU/RAM/POWER/UTILIZATION)
    - Boot MemTest86+ and show memory dianostics running
    - Windows Memory Diagnostic Tool
        - Start MemDT
        - Show scan in progress
    - Utilization
        - Resource monitoring
            - Task Manager and Resource Monitor
            - *top* in Linux

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Memory Diagnostics (memtest86+): https://www.memtest.org/
- Not applicable if blank